AIR-FARE BOOKING SYSTEM

DOCUMENTATION

Submission:

To: Lecturer Miss Ayesha Inayat

For: Object Oriented Programming – Semester Project for the Session Fall 2025 (BSSE-3T)

Dated: 10th January, 2026

Developed by team of 5 members:

Syed Mubazir Haider**,** Abdul Moez**,** Aleeza Bibi**,** Moiz Rizwan**,** Urwa Iqbal

Introduction:

Air transportation has become one of the most important modes of travel in the modern world. With the increase in population, globalization, tourism, and business travel, the demand for efficient airline services has grown rapidly. Traditional airline ticket booking methods relied heavily on manual processes, travel agents, physical counters, and paperwork. These methods were time-consuming, error-prone, and inconvenient for passengers.

The Air-Fare Booking System is designed to overcome these limitations by providing a computerized and automated solution. This system allows passengers to search flights, check seat availability, book tickets, and manage reservations through a digital platform. Airlines can also manage flights, schedules, and passenger data efficiently.

This project focuses on building a simple yet effective Air-Fare Booking System using C++ programming, MySQL database and Python for connecting both, following Object-Oriented Programming (OOP) principles. The system addresses the limitations of traditional manual booking processes, such as human errors, inefficient data management, and slow operations. ABS allow users to register passengers, schedule flights, manage airline and airplane information, book tickets, generate tickets in PDF format, and track flights in real-time.

By combining object-oriented programming principles with modular design, ABS ensure maintainability, scalability, and error handling throughout its operations. This documentation provides a detailed overview of the system architecture, module descriptions, implementation details, user interface, testing, and future enhancements. It serves as both a guide and a reference for users, developers, and evaluators.

Background:

Airline management is a complex process involving multiple tasks such as scheduling flights, managing aircraft, booking tickets, and keeping track of passengers. Traditionally, many airlines relied on manual booking systems, phone-based reservations, or disconnected software solutions. These approaches often lead to inefficiencies, errors, and delays in service delivery.

With the growing demand for efficient airline management systems, there is a need for an integrated software solution that can handle multiple operations simultaneously while providing a user-friendly interface and accurate data management.

Motivation:

The motivation behind developing the ABS project includes:

- Automation of manual processes: Reducing human errors in booking and flight management.
- Centralized data management: Storing airline, flight, passenger, and ticket information efficiently.
- Improved user experience: Simple console-based interface for administrators and users.
- Scalability: Modular design allows adding new features like seat selection, flight reports, or payment integration in the future.

By implementing this system, the team aimed to demonstrate practical application of object-oriented programming in a real-world scenario while creating a functional software that can be extended for larger airline operations.

Scope:

The scope of ABS includes the following functionalities:

- **Airline Management:** Add, view, and manage airline companies including country of origin and airline code.
- **Airplane Management:** Manage aircraft details such as model, seating arrangement, and assignment to airlines.
- **Flight Scheduling:** Schedule flights with source, destination, date, and take-off time.
- **Passenger Management:** Register passengers, maintain contact details, and track last flight.
- **Ticket Booking:** Reserve seats, assign ticket numbers, calculate price, generate PDF tickets, and track booking status.
- **Admin Interface:** Admin login with privilege access to manage airlines, airplanes, and flights.
- **System Environment:** The system is console-based, designed for desktop usage, and integrates Python scripts to handle data persistence in a lightweight database structure.

System Objectives:

The Airfare Booking System is designed with the following objectives:

Functional Objectives

- Register new airlines and airplanes.
- Schedule and manage flights.
- Register passengers and update last flight data.
- Book tickets for passengers on scheduled flights.
- Generate tickets in PDF format for passengers.
- Search flights by location or date
- Display current flights and historical flight data.

Non-Functional Objectives

- Usability: Simple and intuitive console-based interface.
- Reliability: Accurate data management with error handling.
- Scalability: Modular design allows future upgrades.
- Maintainability: Object-oriented design with reusable classes and functions.
- Performance: Efficient execution using C++ and Python scripts for database operations.

System Requirements:

For the purpose of best efficient system use the following pre-requisites are important

Hardware Requirements

- Minimum 4 GB RAM
- Processor: Intel i3 or equivalent
- Hard Disk: 500 MB free space
- Operating System: Windows 10 or later

Software Requirements

- C++ Compiler (Dev C++)
- Python 3.x for database scripts
- Windows API (for console enhancements and icons)
- Optional: PDF library for ticket generation

Development Tools

- **C++:** For core logic and object-oriented design.
- **Python:** For database handling (CRUD operations).
- **SQL:** For database storing data.
- **Windows API:** For console customization and appearance enhancements.
- **Dev C++:** For editing scripts.

System Analysis:

Current System Limitations

Before the ABS, airlines often relied on manual or partial digital systems, which had the following issues:

- **Manual Booking Errors:** Human mistakes during ticket entry or flight scheduling.
- **Inefficient Passenger Tracking: No** centralized record of passengers or flight history.
- **Time-Consuming Operations:** Booking, cancellation, and data retrieval were slow.
- **Data Redundancy:** Repetition of passenger or flight information across multiple files.
- **Limited Access Control:** No role-based admin privileges.

Proposed System Advantages

- **Automation:** Flights, passengers, and tickets are managed through a unified system.
- **Data Accuracy:** Real-time updates reduce errors.
- **User-Friendly Interface:** Console menus with clear options and validations.
- **Modular Design:** Each entity (Passenger, Flight, Airline, Ticket) is independent and reusable.
- **Admin Privileges:** Secure access to critical modules with login authentication.

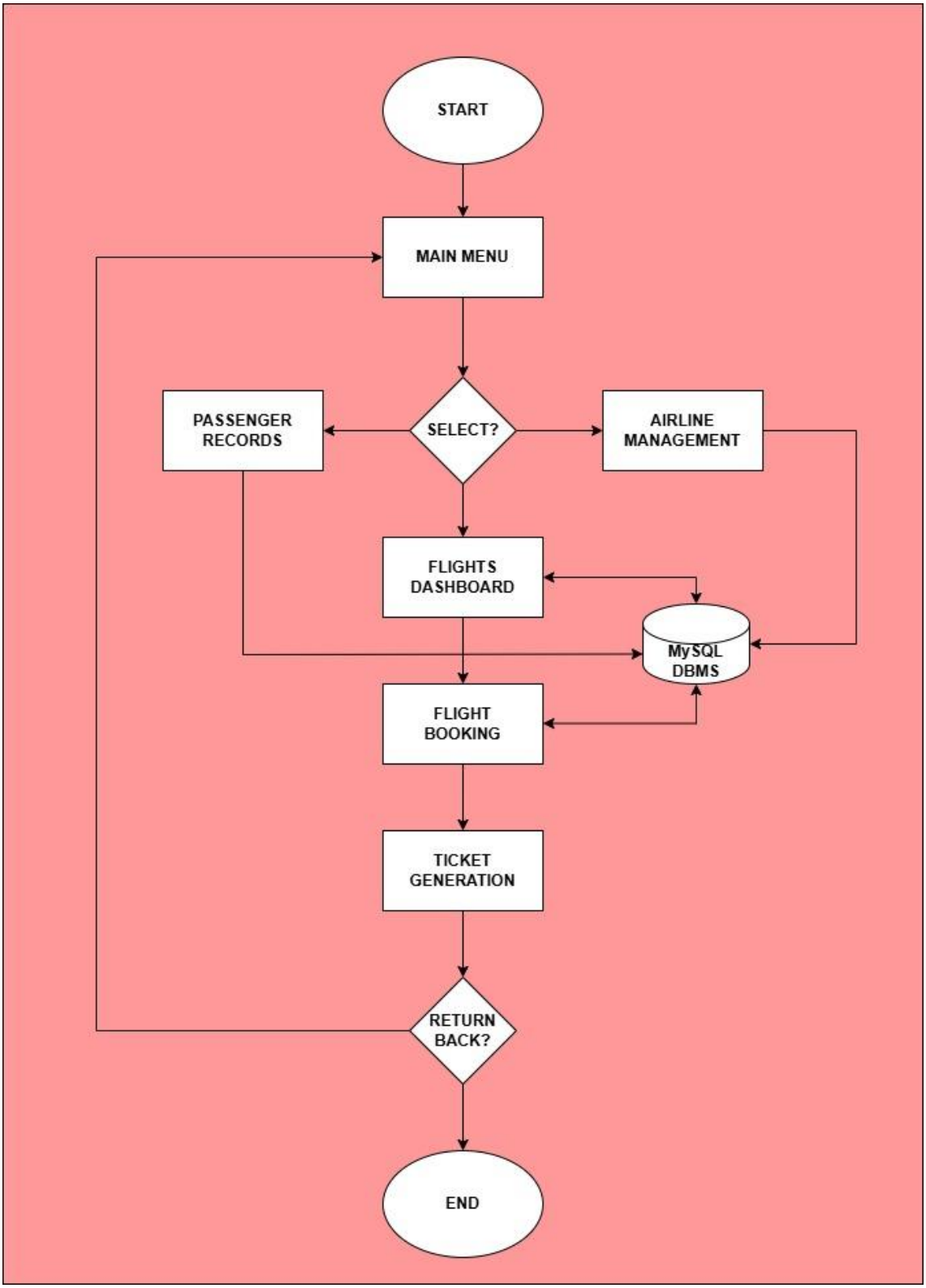
System Design and Architecture:

The Airfare Booking System (ABS) is designed using a **modular object-oriented architecture**, allowing flexibility, maintainability, and scalability. The system is divided into multiple independent modules, each responsible for specific functionalities:

Modules Overview

- **Airline Module** – Manages airline companies and their details.
- **Airplane Module** – Handles aircraft details and assignment to airlines.
- **Flight Module** – Schedules and manages flights with dates, times, and capacity.
- **Passenger Module** – Registers and maintains passenger information.
- **Ticket Module** – Books tickets, generates PDFs, and tracks booking status.
- **Admin Module** – Handles authentication and overall management of the system.

Flowchart of ABS



Classes and Functions:

Airline

Attributes: id, name, country, code

Methods: addAirline(), viewAirline(), updateAirline()

Airplane (inherits from Airline)

Attributes: model, seatCapacity, airlineId

Methods: addAirplane(), viewAirplane(), updateAirplane()

Flight

Attributes: flightId, airplaneId, source, destination, date, time

Methods: scheduleFlight(), viewFlight(), updateFlight()

Passenger

Attributes: passengerId, name, contact, lastFlightId

Methods: registerPassenger(), viewPassenger(), updatePassenger()

Ticket

Attributes: ticketId, flightId, passengerId, seatCode, price, dateOfBooking, status

Methods: bookTicket(), cancelTicket(), generatePDFTicket()

Entity Relationships:

Airline → Airplane: One-to-Many (An airline can have multiple airplanes)

Airplane → Flight: One-to-Many (A plane can be assigned to multiple flights)

Flight → Ticket: One-to-Many (Multiple tickets per flight)

Passenger → Ticket: One-to-Many (Passenger can book multiple tickets)

Airline → Flight: One-to-Many (An airline can have multiple flights)

Airline → Flight: One-to-Many (An airline can have multiple flights)

Module Descriptions:

1.Airline Module

- **Purpose:** Maintain airline details for system operations.
- **Inputs:** Airline name, country, airline code.
- **Outputs:** List of airlines, searchable by name or code.

Operations:

- addAirline(): Adds new airline to database.
- viewAirline(): Displays all registered airlines.
- updateAirline(): Updates airline information.

2.Airplane Module

- **Purpose:** Manage aircraft data, including seat capacity and airline assignment.
- **Inheritance:** Inherits from Airline for airline association.

Operations:

- addAirplane()
- viewAirplane()
- updateAirplane()

Attributes:

- model: Aircraft model.
- seatCapacity: Total seats.
- airlineId: Links to Airline.

3.Flight Module

- **Purpose:** Schedule and manage flights.
- **Key Attributes:** source, destination, date, time, airplaneId.

Operations:

- scheduleFlight(): Adds flight to system.
- viewFlight(): Displays flight schedule.
- updateFlight(): Modify flight details.

Important Logic:

- Ensures no two flights use the same airplane at the same time.
- Checks seat availability before allowing ticket booking.

4. Passenger Module

- Purpose: Store passenger info and manage bookings.
- Attributes: passengerId, name, contact, lastFlightId.

Operations:

- registerPassenger()
- viewPassenger()
- updatePassenger()

5. Ticket Module

- **Purpose:** Book tickets and generate PDF confirmations.
- **Attributes:** ticketId, seatCode, price, status.

Operations:

- bookTicket(): Reserves seat and assigns ticket number.
- cancelTicket(): Cancels booking and updates availability.
- generatePDFTicket(): Produces ticket for passenger.

Important Logic:

- Ensures no double booking of seats.
- Calculates fare based on seat type and flight duration.

6. Admin Module

- Purpose: Control access to modules and maintain system integrity.
- Attributes: username, password.

Workflow:

- Admin inputs data for Airlines, Airplanes, and Flights → System validates and stores.

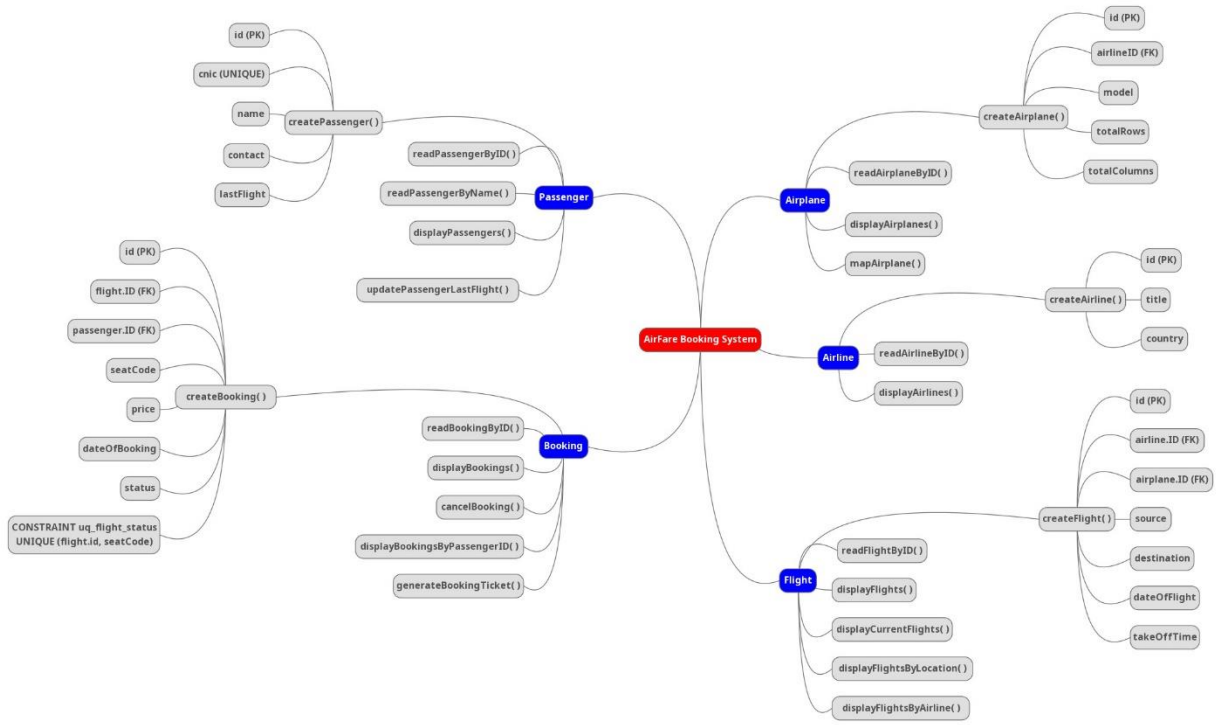
- Passengen Registers → Passenger Module Stores Information.

- Passenger books ticket → Ticket Module verifies flight availability → Updates Ticket Database → Generates Ticket PDF.

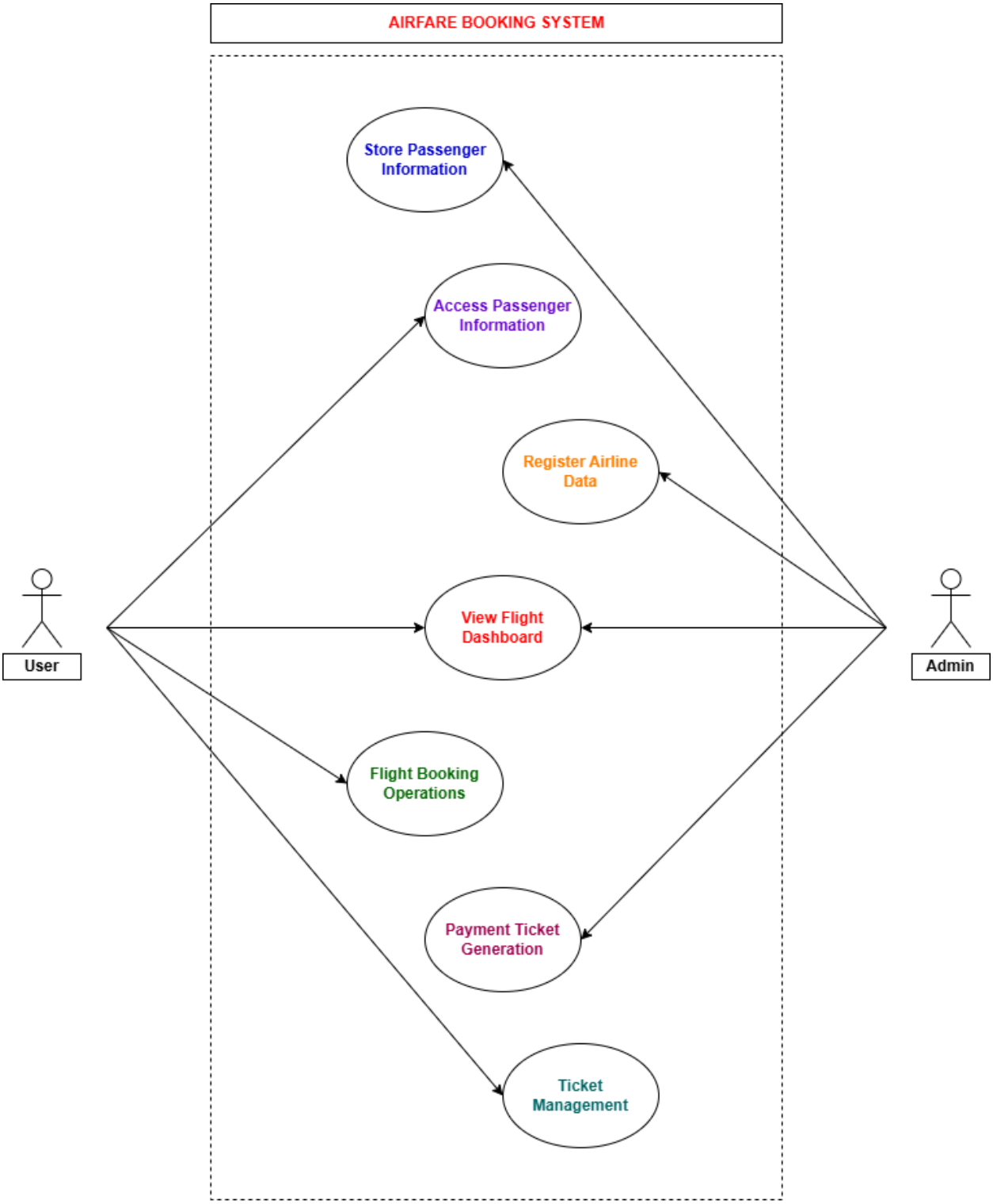
- Flight Module checks for plane availability.

- Ticket Module calculates fare based on seat class.

- Passenger Module updates last flight info after booking confirmation.



Use Case Diagram:



OOP Concepts:

- **Classes and Objects**

  Classes represent real-world entities such as Passenger, Flight, Ticket, Airline, and Airplane.

- **Encapsulation**

  Encapsulation is used to protect sensitive data.

- **Inheritance**

  Different user roles inherit properties from a base User class.

- **Polymorphism**

  Same functions behave differently for different objects.

- **Abstraction**

  Complex processes are hidden from users.

- **Composition and Association**

  Used to define relationships between objects.

- **Constructors and Destructors**

  Used for initialization and cleanup.

Application Interface:

Setup Page

AirFare Booking System - ABS

```
Getting Things Started...
Creating Interface...
Loading: [##################################################] 100%
Successful! Press Enter:
```

Main Page

AirFare Booking System - ABS

```
                              Sat Jan 10 17:37:38 2026
-------------------------------------------------------------------------------
|                                                                             |
|                                                                             |
|                              AIR-FARE BOOKING SYSTEM                        |
|                                                                             |
|                                                                             |
-------------------------------------------------------------------------------

                                     MAIN MENU
 Type @ -> Admin                                                 Type e -> Exit
-------------------------------------------------------------------------------

                                1. Flights Dashboard

                                2. Passenger Records

                                3. Airline Information
```

Flight Dashboard

```
AirFare Booking System - ABS
--------------------------------------------------------------------------------
|                                                                              |
|                                                                              |
|                              FLIGHTS DASHBOARD                               |
|                                                                              |
|                                                                              |
--------------------------------------------------------------------------------

                        1. Display Available Flights

                        2. Search Flights by Location

                        3. Display All Flights Record

                        0. Return to Main Menu

Enter ->
```

Passenger Records

```
AirFare Booking System - ABS
--------------------------------------------------------------------------------
|                                                                              |
|                                                                              |
|                              PASSENGERS RECORD                               |
|                                                                              |
|                                                                              |
--------------------------------------------------------------------------------

                        1. Enter New Passenger Info

                        2. Search Passenger by ID#

                        3. Search Passenger by Name

                        4. Display All Passengers Records

                        5. Manage Passenger Tickets

                        0. Return to Main Menu

Enter ->
```

Login Page

AirFare Booking System - ABS

```
Airline Module
                        ------------------------------------------------------------
                        -----------------  AirFare Booking System  -------------------
                        ------------------------------------------------------------
                        ------------------------- LOGIN PAGE --------------------------
                        ------------------------------------------------------------
                        Enter your Credentials :

                        Username : admin
                        Password : ****

                        Login Successful
```

Airline Administration

AirFare Booking System - ABS

```
--------------------------------------------------------------------------------
|                                                                              |
|                                                                              |
|                              AIRLINE ADMINISTRATION                          |
|                                                                              |
|                                                                              |
--------------------------------------------------------------------------------

                        1. New Airline Info

                        2. New Airplane Info

                        3. Show Airline/Airplane Info

                        4. Display All Airline Records

                        5. Manage Airline Flights

                        0. Return to Main Menu

Enter ->
```

Add Flight Info

AirFare Booking System - ABS

```
Adding New Flight

All Airlines:

Airline ID              Title                Country
-------------------------------------------------------
    1                  Emirates               UAE
-------------------------------------------------------
    2                   SkyAir                USA
-------------------------------------------------------
    3                    PIA                  PAK
-------------------------------------------------------
    4                  AirBlue                Pakistan
-------------------------------------------------------
    5              British Airways            UK
-------------------------------------------------------
    6               Qatar Airways             Qatar
-------------------------------------------------------
    7               Serena Airways            Pakistan
-------------------------------------------------------
    8               Eagles Airline            Portugal
-------------------------------------------------------
Continue: Enter Airline ID: #1
Airline Found:
-------------------------------------------------------
AirlineID       : 1
Airline         : Emirates
Country         : UAE
Flight Code     : EMR
-------------------------------------------------------
Continue:
```

Add Flight Information

AirFare Booking System - ABS

```
All Airplanes:

Airplane ID     Airline ID         Model Name         Rows        Columns
-----------------------------------------------------------------------------
  1                1               Boeng-707          12            6
-----------------------------------------------------------------------------
  2                1               Boeng-705          14            6
-----------------------------------------------------------------------------
  3                1               Boeng-703          16            6
-----------------------------------------------------------------------------
  4                1                Hello123          14            6
-----------------------------------------------------------------------------
  5                1                Mello456          10            5
-----------------------------------------------------------------------------
Enter Plane ID: #1
Enter Source City Code:         LHR
Enter Destination City Code:    KHI
Enter Date Of Flight:
Enter Day (1 - 31)      :12
Enter Month (1 - 12):   1
Enter Year (>=2026):    2026
Enter Take Off Time:
Enter Hour (0 - 23): 19
Enter Minute (00/15/30/45): 00
Flight record inserted successfully!
Continue:
```

Flights Display

AirFare Booking System - ABS

```
Displaying Available Flights

Flights for Today:


            Flight ID      : 11
            Airline ID     : 3
            Airplane ID    : 6
            Source         : KHI
            Destination    : LHR
            Date of Flight : 10/01/2026
            Take Off Time  : 13:00
            -------------------------


            Flight ID      : 12
            Airline ID     : 2
            Airplane ID    : 11
            Source         : KHI
            Destination    : JFK
            Date of Flight : 10/01/2026
            Take Off Time  : 17:00
            -------------------------


            Flight ID      : 13
            Airline ID     : 2
            Airplane ID    : 11
            Source         : JFK
            Destination    : LHR
            Date of Flight : 10/01/2026
            Take Off Time  : 19:00
            -------------------------


            Flight ID      : 14
            Airline ID     : 4
            Airplane ID    : 12
            Source         : LHR
            Destination    : RYK
            Date of Flight : 10/01/2026
            Take Off Time  : 16:00
            -------------------------
```

Booking Page

AirFare Booking System - ABS

```
--------------------------------------------------------------------------------
Flight-ID#:             12
Airline-ID#:            2
Airplane-ID#:           11
Source:                 KHI
Destination             JFK
Date Of Flight:         10/01/2026
Take Off Time           17:00
--------------------------------------------------------------------------------
Flight ID # 12
--------------------------------------------------------------------------------

                        B. Book Ticket Reservation

                        M. Manage Ticket Bookings

                        R. Return to Main Menu

Enter -> b
Enter Passenger-CNIC:   #3520279715375
Passenger already exists.
FIND
ID: 1
CNIC: 3520279715375
Name: Syed Mubazir Haider
Contact: 03186528350
Last Flight: 10/01/2026

LETS BOOK TICKET...
Enter Passenger-ID: #1
```

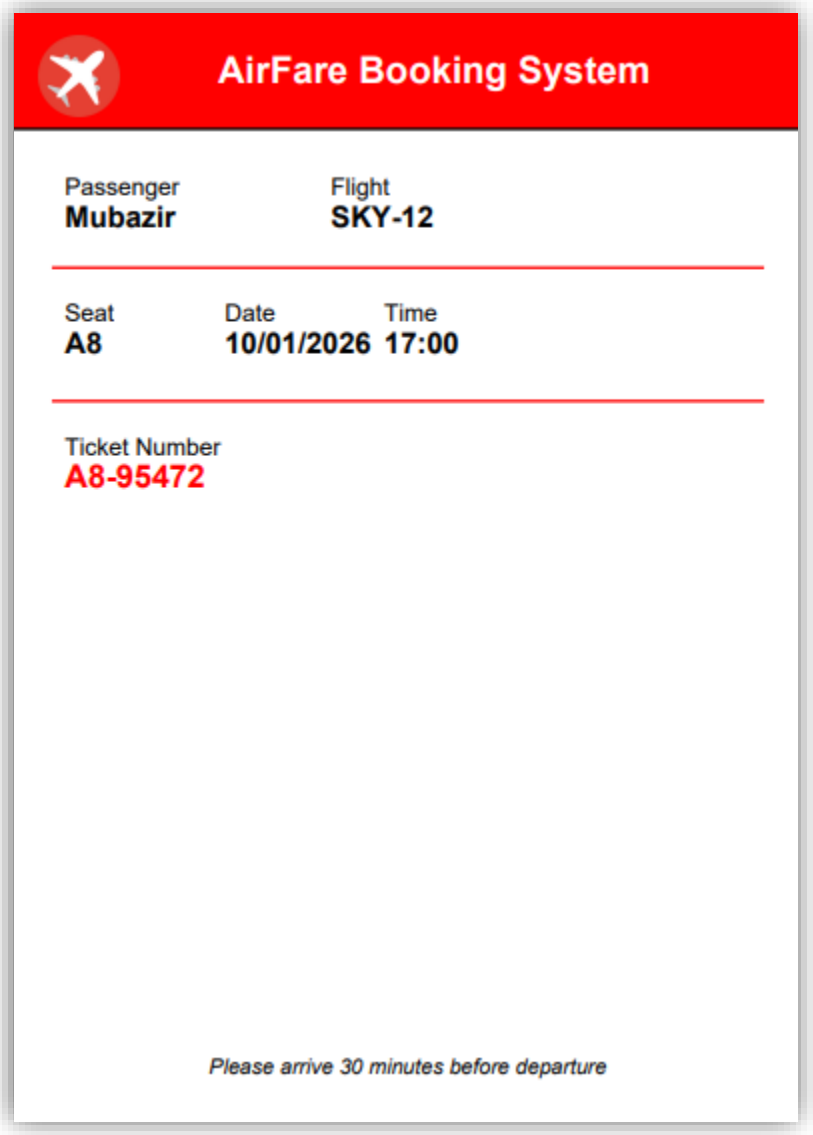
Seat Mapping Selection

```
AirFare Booking System - ABS
Select Seat on Airplane:
-------------------------------------------------------------------------------------------------------------
|      A1      ||      A2      ||      A3      ||      A4      ||      A5      ||      A6      ||      A7      ||      A8      |
-------------------------------------------------------------------------------------------------------------
|      B1      ||      B2      ||      B3      ||      B4      ||      B5      ||      B6      ||      B7      ||      B8      |
-------------------------------------------------------------------------------------------------------------
|      C1      ||      C2      ||      C3      ||      C4      ||      C5      ||      C6      ||      C7      ||      C8      |
-------------------------------------------------------------------------------------------------------------
|      D1      ||      D2      ||      D3      ||      D4      ||      D5      ||      D6      ||      D7      ||      D8      |
-------------------------------------------------------------------------------------------------------------
|      E1      ||      E2      ||      E3      ||      E4      ||      E5      ||      E6      ||      E7      ||      E8      |
-------------------------------------------------------------------------------------------------------------
|      F1      ||      F2      ||      F3      ||      F4      ||      F5      ||      F6      ||      F7      ||      F8      |
-------------------------------------------------------------------------------------------------------------
|      G1      ||      G2      ||      G3      ||      G4      ||      G5      ||      G6      ||      G7      ||      G8      |
-------------------------------------------------------------------------------------------------------------
|      H1      ||      H2      ||      H3      ||      H4      ||      H5      ||      H6      ||      H7      ||      H8      |
-------------------------------------------------------------------------------------------------------------
|      I1      ||      I2      ||      I3      ||      I4      ||      I5      ||      I6      ||      I7      ||      I8      |
-------------------------------------------------------------------------------------------------------------
|      J1      ||      J2      ||      J3      ||      J4      ||      J5      ||      J6      ||      J7      ||      J8      |
-------------------------------------------------------------------------------------------------------------
|      K1      ||      K2      ||      K3      ||      K4      ||      K5      ||      K6      ||      K7      ||      K8      |
-------------------------------------------------------------------------------------------------------------
|      L1      ||      L2      ||      L3      ||      L4      ||      L5      ||      L6      ||      L7      ||      L8      |
-------------------------------------------------------------------------------------------------------------
|      M1      ||      M2      ||      M3      ||      M4      ||      M5      ||      M6      ||      M7      ||      M8      |
-------------------------------------------------------------------------------------------------------------
|      N1      ||      N2      ||      N3      ||      N4      ||      N5      ||      N6      ||      N7      ||      N8      |
-------------------------------------------------------------------------------------------------------------
|      O1      ||      O2      ||      O3      ||      O4      ||      O5      ||      O6      ||      O7      ||      O8      |
-------------------------------------------------------------------------------------------------------------
|      P1      ||      P2      ||      P3      ||      P4      ||      P5      ||      P6      ||      P7      ||      P8      |
-------------------------------------------------------------------------------------------------------------

Enter Seat Code (eg. A12): A8
Enter Ticket Price: 5600
```

Ticket Generation

```
AirFare Booking System - ABS
Opening Ticket Module with Create Authority

Generating Ticket for Passenger #1
-----------------------------------------------------------
FLIGHT ID:                #12
SEAT CODE:                A8
BOOKING PRICE:            5600
DATE OF BOOKING:          10/01/2026
BOOKING STATUS:           1
-----------------------------------------------------------
Press 1 to Create: 1
Ticket record inserted successfully!
Continue:
Generate Ticket:
Please Write Name for Ticket: Mubazir
Ticket Generated: AirFare_Ticket_A8-95472.pdf
Ticket PDF generated successfully.
```

Ticket PDF Created



AirFare Booking System

Passenger
Mubazir

Flight
SKY-12

Seat
A8

Date
10/01/2026

Time
17:00

Ticket Number
A8-95472

Please arrive 30 minutes before departure

END OF DOCUMENTATION

PAGE 22 of 22